

Team Name: Elite Coders

Team Members:

Monisha S (240701335)

Pavithra B (240701380)

Devishree J (240701702)

Project Name: Jet Cultivation

Experiment 5: Sequence Diagram

Procedure: Creating Sequence Diagram in draw.io and Uploading to Azure DevOps

Step 1: Open draw.io (diagrams.net)

1. Open browser and go to: <https://app.diagrams.net>
2. Choose Device / Google Drive
3. Click Create New Diagram

Step 2: Create New Diagram

1. Enter file name (e.g., *Sequence Diagram*)
2. Select Blank Diagram
3. Click Create

Step 3: Add Actors and Objects

1. From left panel, go to UML → Actor
2. Add:
 - Farmer
3. Add Object/Lifeline boxes:
 - Web App
 - Auth Service
 - River Service
 - Irrigation Service
 - DB

Step 4: Add Lifelines

1. Draw vertical dashed lines below each object
2. These represent the time flow

Step 5: Add Messages (Arrows)

Login Process

- Farmer → Web App : Login
- Web App → Auth Service : Validate Credentials
- Auth Service → DB : Check User
- DB → Auth Service : Result

Step 6: Add Conditional Flow (alt block)

1. Draw a frame (rectangle)
2. Label it:
3. alt [Login Success]

Inside it:

- Auth Service → Web App : Success
- Web App → River Service : Get River Status
- River Service → DB : Fetch Data
- DB → River Service : River Status
- River Service → Web App : Status

Step 7: Add Nested Condition

Inside previous block:

1. Add another alt block

alt [River Good]

- Web App → Irrigation Service : Primary Irrigation

[River Not Good]

- Web App → Irrigation Service : Alternative Irrigation

Step 8: Continue Flow

- Web App → Farmer : Show Dashboard

Step 9: Add Failure Condition

Outside main success block:

[Login Failed]

- Auth Service → Web App : Failed
- Web App → Farmer : Error Message

Step 10: Format Diagram

1. Align lifelines properly
2. Keep arrows straight
3. Use dashed lines for return messages
4. Ensure boxes cover full flow

Step 11: Save Diagram

1. Click File → Save As (.drawio)
2. Export as:
 - PNG / JPG

Step 12: Open Azure DevOps

1. Go to your project
2. Navigate to:
3. Wiki or Repos

Step 13: Upload Diagram

Wiki:

1. Open page
2. Click Edit
3. Insert Image
4. Upload diagram

Repos:

1. Click Upload Files
2. Add image
3. Commit

Step 14: Verify

1. Open uploaded diagram
2. Check alignment and clarity

Result

The Sequence Diagram for Jet Cultivation System is successfully created using draw.io and uploaded to Azure DevOps.

